

The eMule application of the eDonkey protocol



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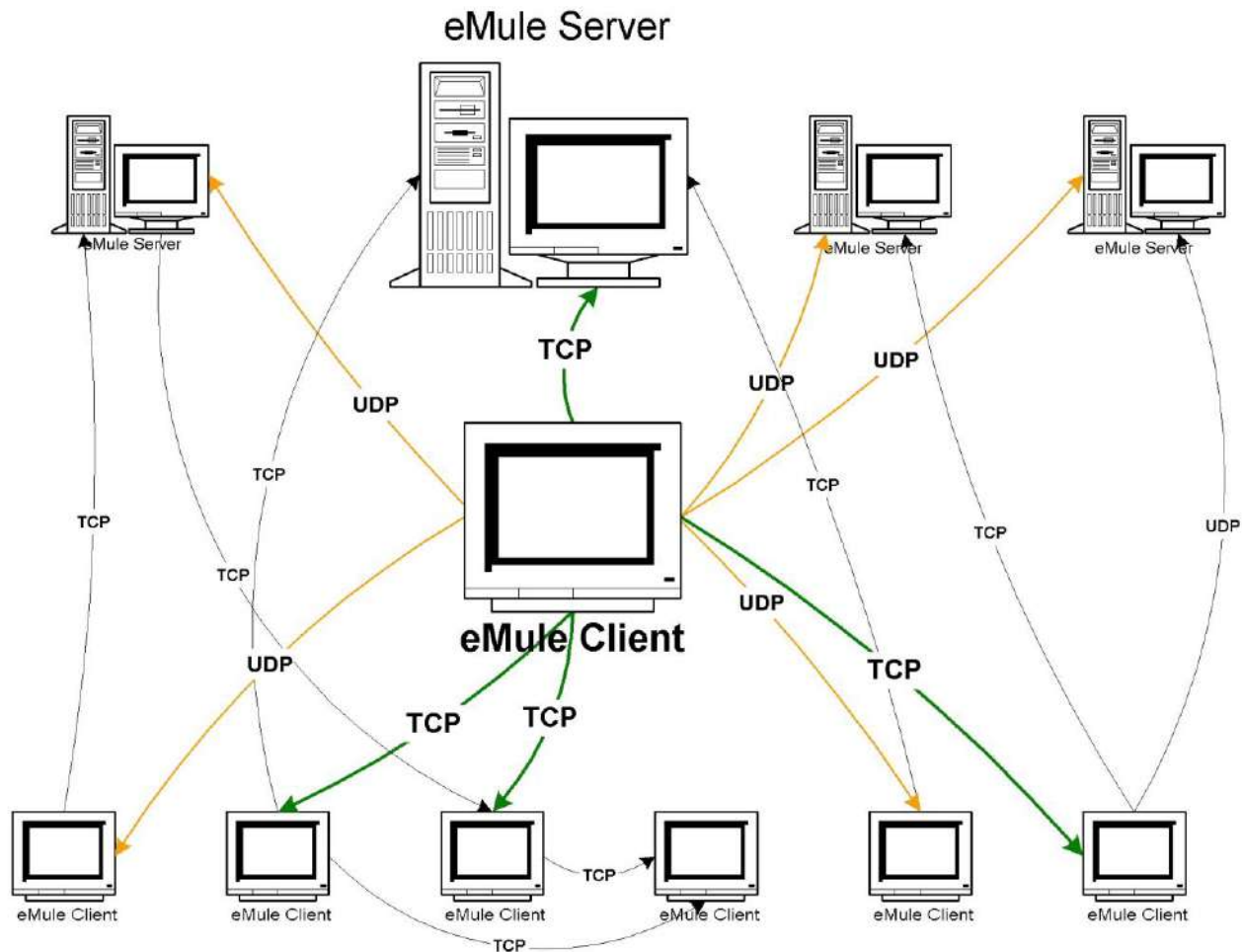
Agenda

- Introduction
- The protocol
 - Client - Server protocol
 - Client to Client protocol
- The application
 - Queue management
 - Credit system
 - Download part ordering
 - Corrupted download recovery

Introduction

- eMule is a P2P file sharing network
 - Based on an extended eDonkey protocol
 - Uses both TCP and UDP
 - Client Server architecture
- Open source client
 - High rate of expansions and improvements
 - Latest sources can be found at [sourceforce](#)
- There is also an official [website](#)

The eMule Network



Terminology

- **Client ID** – 32 bit identifier provided by the server upon connection
 - Valid only through the session lifetime.
 - May be either a high ID or a low ID
- **User ID** – A 128 bit value produced by concatenating random numbers
 - Used to identify a client across sessions
- **File ID** – A 128 bit hash calculated using MD4.
 - A file is divided to 9.28 MB parts which are hashed separately and then combined

The Protocol

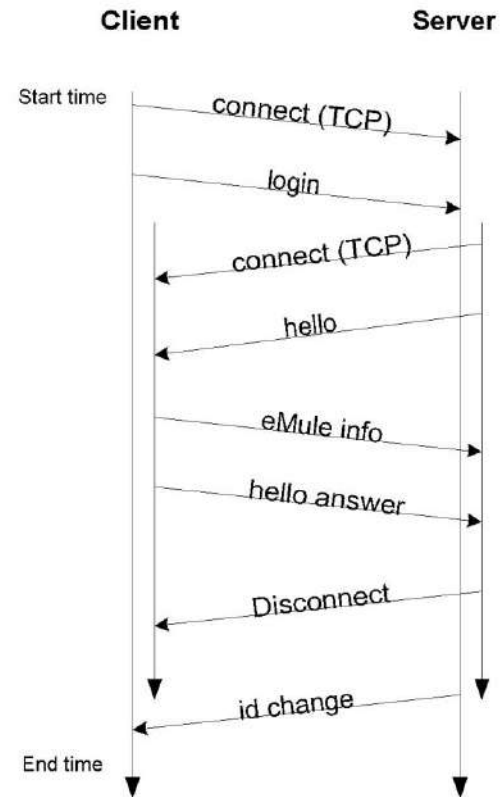
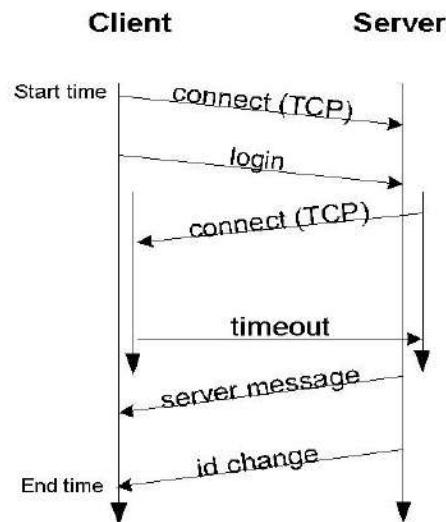
- Each message starts with a 6 byte header [protocol, size ,type]
 - Empty messages are allowed
- Encoded are Integers, Strings and bit sets
 - Strings are encoded using a 2 byte length, value encoding
- Message tags

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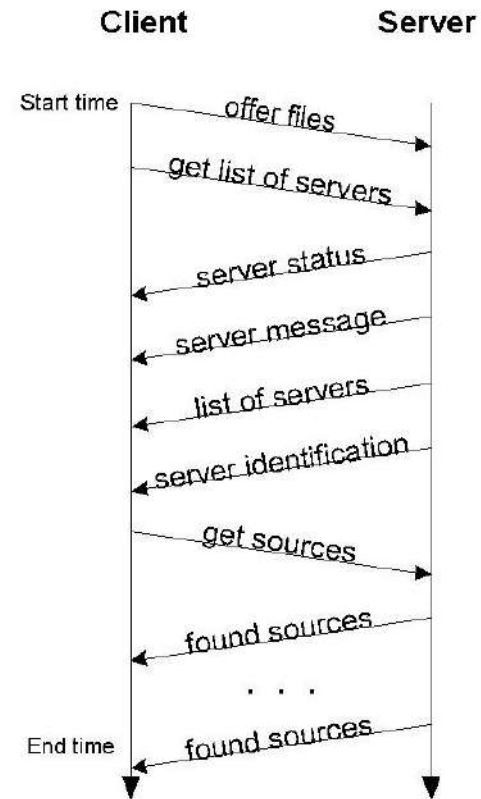
Connection establishment

- Client Identification
- Determines whether the client gets a low ID



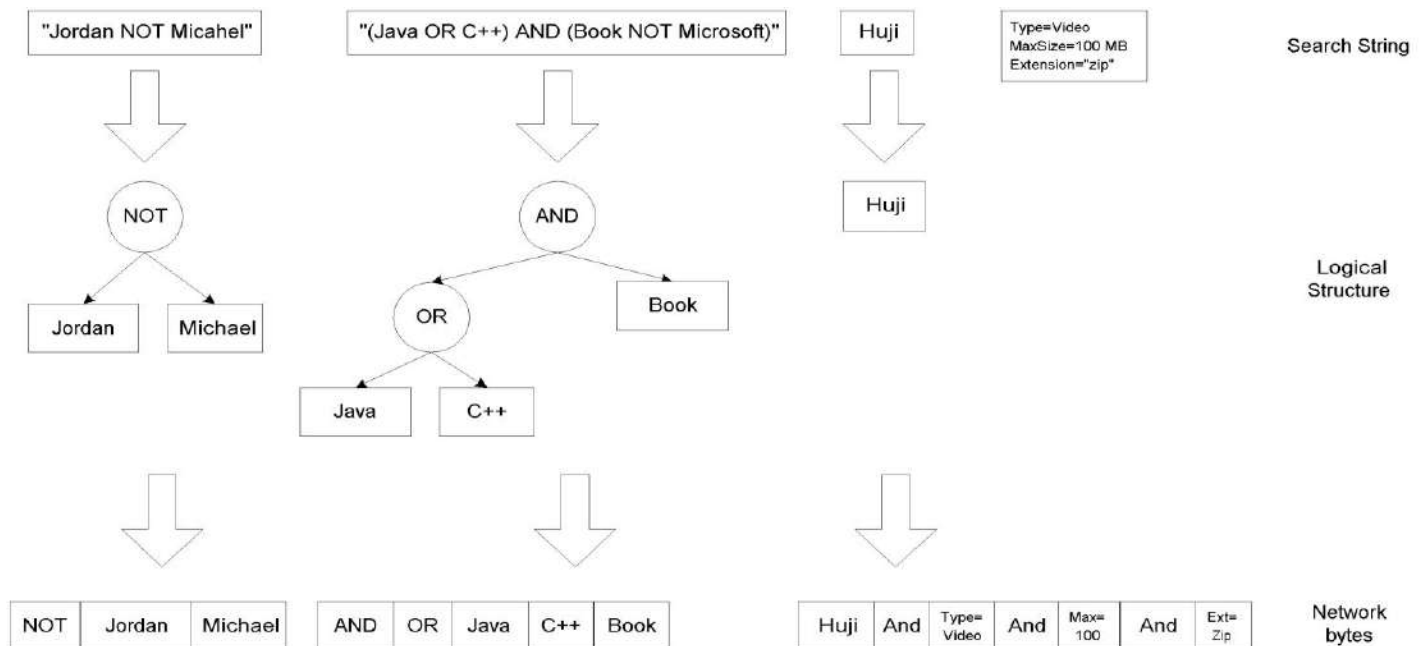
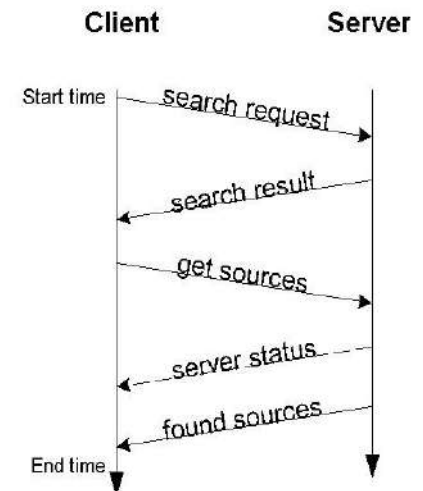
Connection startup

- This scenario assumes that the client has a requested file list
- The client sends his files
- The server replies with status information
- The client asks for sources for his files
- The server replies with his available sources



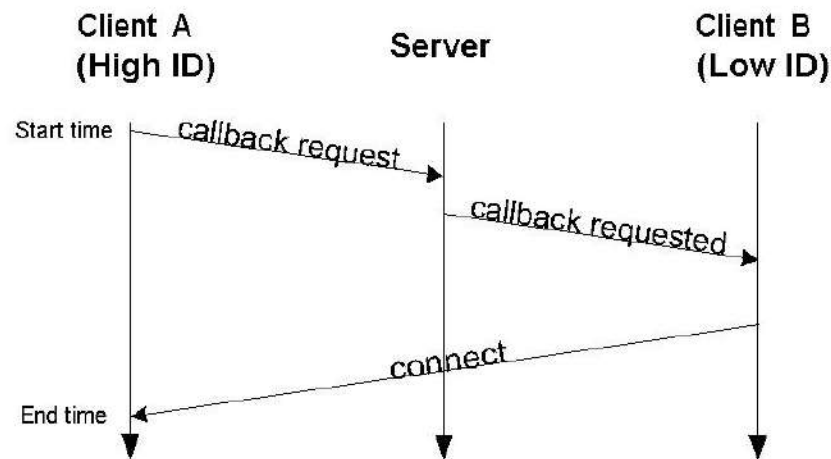
File search

- Search packet supports Boolean conditions
- A search usually follows by a source request



Callback mechanism

- Allows a high ID client to connect to a low ID client by asking a call back
- Both clients must be connected to the same server

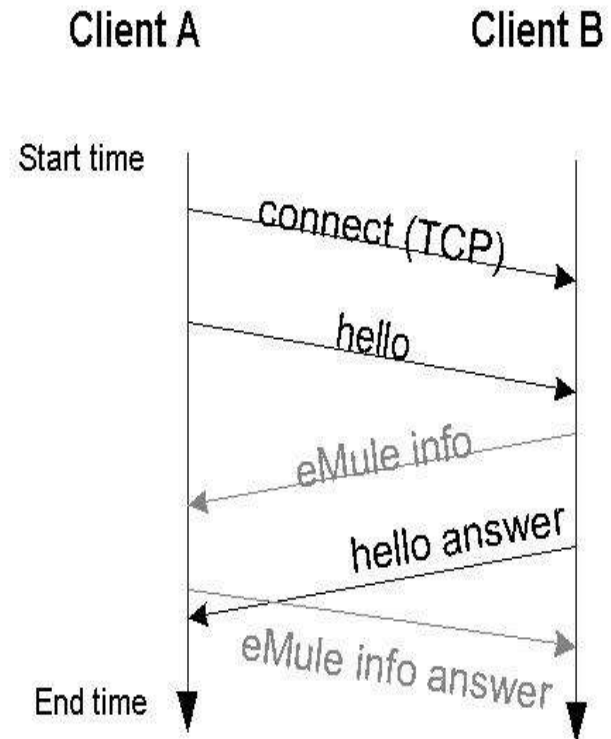


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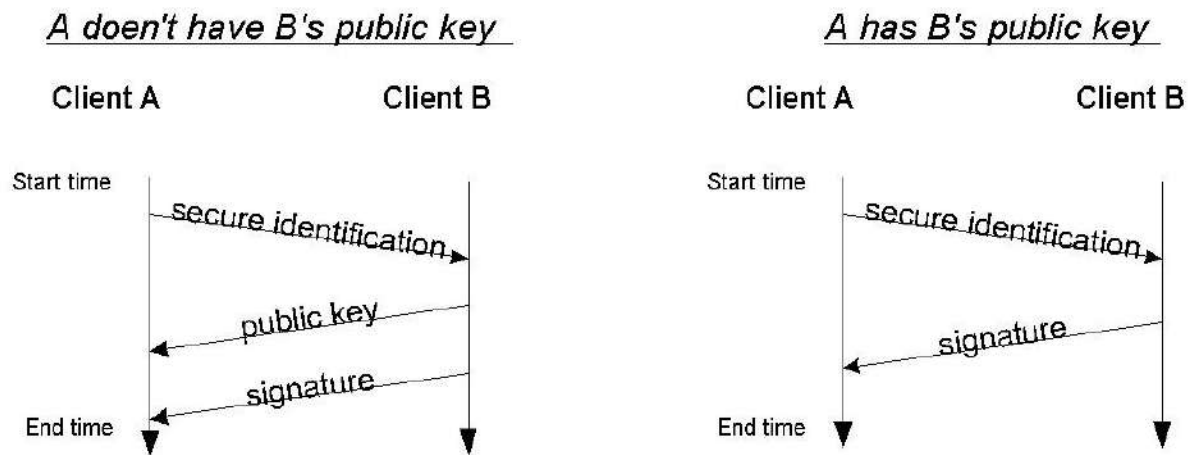
Initial Handshake

- The handshake is symmetric
- The eMule info message is used mainly to determine peer capabilities
- Messages marked in gray are part of the eMule protocol extensions



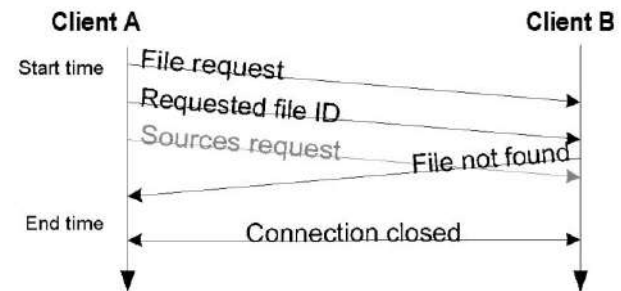
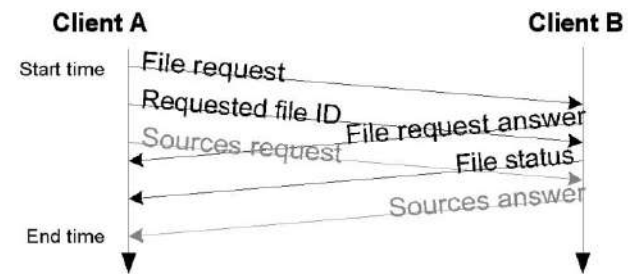
Secure identification

- Used to prevent user impersonation motivated by the credit system
- Defined in the extended protocol, RSA based, simple challenge response scheme



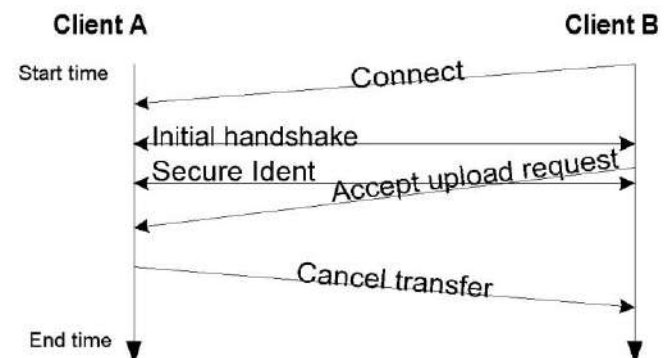
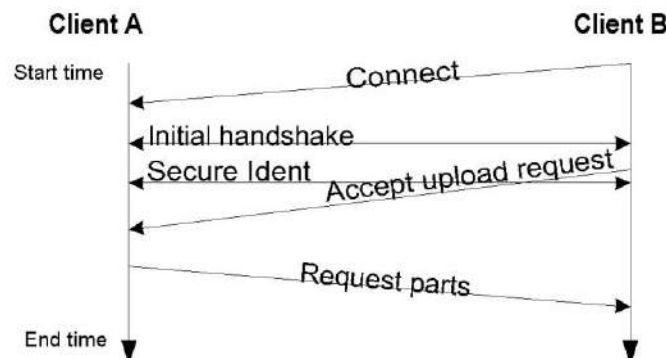
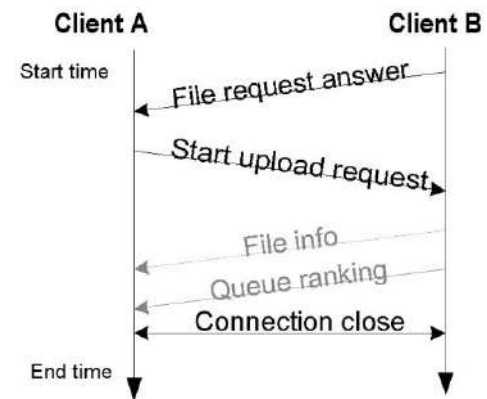
File requests

- There is redundancy in the messages
- The sources request and answer complement the server source query and is useful when the other client is still downloading the file



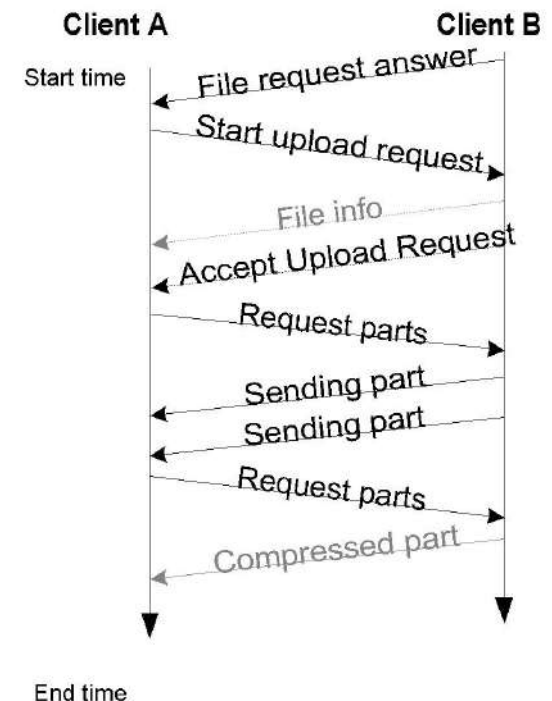
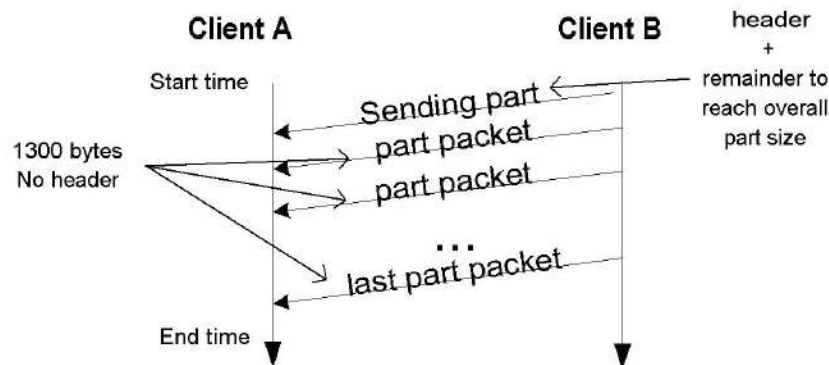
Enlisting in the upload queue

- When A's queue is not empty B is enlisted into A's queue
- When B reaches the top of A's queue, A connects to B in order to start the upload



Data Transfer

- The major part of eMule's network activity
- The extended protocol allows sending compressed parts



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Download queue

- The eMule client connects to sources in the order they were received from the server / UDP connection
- There is a special mechanism which deals with the corner case where the remote client is a potential source for several files

Upload queue

- A priority queue, based on waiting time
- At the head of the queue are clients with the highest **Score**
 - $\text{Score} = \text{Rating} \times (\text{seconds in queue})/100$
 - Or 0xFFFFFFFF (infinite) in case of a friend
- The rating initial value is 100
- Banned users rating value is 0

Upload queue, cont

- Downloading clients initial rating is different
 - In the first 15 minutes of a download the initial rating is 200
 - A new downloading client with higher rating may push a downloading client out
- The rating is modified according to these factors
 - Credits (modifier $x1 - x10$)
 - Shared file priority (modifier $x0.2 - x1.8$)

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The credit system

- The credit system awards users for uploading files
- Client credits are not global, it is a value assigned to every ordered (client, client) pair
- The credit values range between 1-10
- Secure identification prevents false credit claims

Calculating the credit value

- The minimum of the following
 - $\text{Uploaded Total} \times 2 / \text{Downloaded Total}$
 - $\text{SQRT}(\text{Uploaded Total} + 2)$
- Boundary conditions:
 - If Uploaded Total is less than 1MB then Modifier = 1
 - If Downloaded Total = 0 then Modifier = 10
 - The modifier may not be lower than 1 or higher than 10

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Selecting parts to download

- A file is divided to 9.28 MB parts
- Each part is divided to 180 KB blocks
- The downloading client determines which parts are downloaded
- All the blocks that are requested from the same source reside in the same part

Selecting parts, cont.

- A part is selected as follows:
 1. Rare parts are downloaded first
 2. Preview parts (first and last)
 3. Spread requests among sources
 4. Complete partially received parts
- The principles impose part rating:
 1. 0-9999 unrequested and requested very rare parts
 2. 10000-19999 unrequested rare and preview parts
 3. 20000-29999 unrequested most complete common part
 4. 30000-39999 requested rare and preview parts
 5. 40000-49999 requested uncompleted common parts
- Lowest rated part is downloaded first

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Basic corruption handling

- The default corruption handling mechanism named ICH
- The algorithm:
 - When a part download completes, compute and compare its hash against the source
 - In case of a mismatch, blocks are downloaded one by one until the corruption is corrected

Advanced corruption handling

- A new feature (since 2004) also called AICH
- Seems that it is soon to be replaced
- A tree based recovery mechanism

Backup slides

- Screen shots
- Client server UDP communication
- Client to Client UDP communication

The End

Thank you

Servers screen

eMule v0.30e

Disconnect Servers Transfers Search Shared Files Messages IRC Statistics Preferences

Servers (122)

Server name	IP	Description	Ping	Users	Max ...	Files
Razorback 2	195.245.244.243 : 4661	www.razorback2.com / d...	1187	413642	900000	4520309
DonkeyServer No1	62.241.53.2 : 4242	www.Use-Next.de / dual ...	203	272897	850000	2958724
!StormSex.net - Free Porn!	69.50.187.210 : 4661	www.stormsex.net	422	97344	200000	1725100
DonkeyServer No2	62.241.53.16 : 4242	www.Use-Next.de	297	126246	350000	1458454
### SauGelleGirls ###	81.23.250.167 : 4242	www.saugelegirls.de	110	160934	161161	1433092
DonkeyServer No6	62.241.53.4 : 4242	www.Use-Next.de	281	106799	200000	1298922
DonkeyServer No3	62.241.53.17 : 4242	www.Use-Next.de	156	88754	350000	1091621
### Porno-Flatrate ###	81.23.250.169 : 4242	www.porno-flatrate.net	672	100545	111111	962072
66.192.117.38	66.192.117.38 : 4661		328	74755	80000	919486
DonkeyServer No4	62.241.53.15 : 4242	www.Use-Next.de	422	67908	200000	840193
Project Grande	213.158.119.104 : 4661	Project Grande	422	81635	115000	832211
Sci-Fi-Network Server	69.50.228.50 : 4646	http://scifi-scene.dyndns...	500	117185	222222	815047
gAi@EeAEIEEeAEu@8gA http://www.kd...	211.214.161.107 : 4661	gAi@EeAEIEEeAEu@8gA ...	891	38831	45000	749631
66.192.117.42	66.192.117.42 : 4661		359	58143	60000	715158
### DOINGDO ###	205.209.136.190 : 4661	www.webspacevermietu...	688	18176	100000	708458
38.119.67.93	38.119.67.93 : 4661		343	56987	60000	700940
TUF Taiwan ED Server	140.123.108.139 : 7654	TUF Taiwan ED Server	438	39700	40000	691100
Breizh Digitalis	213.186.47.84 : 4661	http://www.nana-sexy.com	359	66614	100000	629170

Add Server

IP or Address: Port: 4661

Name:

Add

Update server.met from URL

Update

My Info

Status: Connected
IP:Port: 82.81.90.196:14374
ID: 3294253394
High ID
WebIfc: Disabled

Reset

ServerInfo Log Verbose

Economisez 6 ou 15 EUROS sur votre prochain achat de cd vierges ou sur un autre produit : <http://www.razorback2.com/?pageid=offres>
Des vacances à prix cassés : <http://www.razorback2.com/?pageid=offres2>
Voyance et tarots GRATUITS : <http://www.razorback2.com/?pageid=offres3>

----- CONCOURS EXCEPTIONNELS -----
Gagnez UNE PEUGEOT 206 : <http://www.ed2k.ch/razorback2/?pageid=concours>
Gagnez 2 TELEPHONES portables Samsung dernier model SGH E-800 : <http://www.razorback2.com/?pageid=concours>
Gagnez un téléphone portable Nokia 7250 I : <http://www.razorback2.com/?pageid=concours>
Gagnez une TV 16/9ème, des lecteurs de dvd, des places de ciné : <http://www.razorback2.com/?pageid=concours>

Soutenez notre serveur, c'est TRES IMPORTANT / Please visit our site, it's VERY IMPORTANT
Besucht bitte unsere Seite, Es ist sehr Wichtig !!
<http://www.razorback2.com/pub.html>

Connection established on:Razorback 2

Users: 412K(2.73M) | Files: 45.01M(326.05M) | Up: 9.7 | Down: 0.0

Razorback 2

Search screen

eMule v0.30e

Disconnect Servers Transfers Search Shared Files Messages JRC Statistics Preferences

Search
Name: Peer to peer
Type: Any Method: Global (Server) [Reset]

Filter
Min size: MB Max size: MB
Availability: Extension:

Direct Download
eD2K Link(s):

[Start] [More] [Cancel]

Search Results
[X] Infected Mushroom (200) [X] Peer to peer (114) [Clear All]

File Name	Size	Availabl...	Type	FileID
Connecta.2000.v.5.20.1.(audio.videoconferencia.peer.to.peer).exe	5.71 MB	58 (1)	Programs	28CC589D75362
Peer Guardian 1.99pr4 para ocultar las descargar P2P (spanish español castellano matrix tomb discografia hulk gta autocad).rar	809 KB	55 (1)	Archive	D84C5D8CEE7a
Steganos Peer To Peer p2P Securise v6.0.4 French-FF.rar	7.15 MB	46 (1)	Archive	6167464271D79
Connecta.2000.v.5.30.1.(audio.videoconferencia.peer.to.peer).exe	5.79 MB	38 (1)	Programs	AF08E4E426B80
Connecta.2000.v.5.20.0.(audio.videoconferencia.peer.to.peer).exe	5.70 MB	32 (1)	Programs	0C3821B78D7E0
Warning - Attention - (English And French Information) - Peer To Peer Enemies - Peer2Peer - p2P - Peerguardian.bit	73 KB	25 (1)	Any	9CB01812F3EA6
peer.guardian.lst,USE.THIS.TO.PROTECT.YOURSELF,.IPFILTER,.DOWNLOAD.NOW.UPDATED,AS.OF.21-04-2003.bit	137 KB	19 (1)	Any	E96C3491C5316
Peer to Peer- Harnessing the Power of Disruptive Technologies.pdf	2.12 MB	18 (1)	Any	CA86366A267Cf
Peer Guardian1.98.1 para ocultar las descargas P2P (spanish español castellano matrix tomb discog.rar	797 KB	17 (1)	Archive	50A57CB8626Cf
Kazaa Lite 2.4.3 (K++ Edition) - english - File sharing - Peer to Peer-No spyware-Sept 25-2003.zip	6.08 MB	16	Archive	BD12E4A80217F
P2P Peer-To-Peer Architecture For Distributed Knowledge Management.ppt	93 KB	13 (1)	Any	858A272B0C9C
Kademlia - A Peer to Peer Information System.pdf	79 KB	13 (1)	Any	B96817E90CDD4
[eBook] P2P - Peer-to-Peer Computing.pdf	524 KB	10 (1)	Any	7D5087E4CF483
Wiley - Gradecki - Mastering JXTA- Building Peer-to-Peer Applications.pdf	2.46 MB	9 (1)	Any	A3DD052D59E9f
Kazaa Lite 2.4.2 (K++ Edition) - english - File sharing - Peer to Peer-No spyware-August 28-2003.zip	5.06 MB	9	Archive	5B7E18689F2F7
How to get faster and cleaner downloads from peer to peer groups, Kazaa,Morpheus,Shareaza,etc.rtf	5 KB	9 (1)	Any	CEFA3D8AC81A
Peer-To-Peer Grid Databases For Web Service Discovery (2002).pdf	815 KB	8 (1)	Any	0188FC2316842
Convergence Of Peer-To-Peer And Grid Computing.pdf	62 KB	8 (1)	Any	E1C6F9A978846
Apress,.Peer.to.Peer.with.VB.Dot.NET.(2003).LIB.chm	7.82 MB	8 (1)	Any	F23C5F8D1D13F
(eBook - pdf) Networking Peer to Peer Manual.pdf	1.03 MB	7 (1)	Any	596529D20D4DE
Hungry Minds - Peer-to-Peer Development and Production.pdf	2.40 MB	6 (1)	Any	17AD3CBA58A1f
LimeWire Pro 3.3.5 Win(Multilingual) Full Retail- File sharing - Peer to Peer (1).zip	13.52 MB	6 (1)	Archive	FF089EEBC4EFD
Shareaza - How to get faster and cleaner downloads from peer to peer groups, Kazaa,Morpheus,Shareaza,etc.rtf	2 KB	5 (1)	Any	62ADE9FBAC61f

[Download Selected]

Connection established on:Razorback 2 Users: 430K(3.07M) | Files: 47.43M/337.32M Up: 9.7 | Down: 0.0 Razorback 2

Download screen

eMule v0.30e

Disconnect Servers Transfers Search Shared Files Messages IRC Statistics Preferences

Downloads (11)

File Name	Size	Transferred	Completed	Speed	Progress	Sources	Priority	Status
Kademlia - A Peer to Peer Information System.pdf	79 KB	0 Bytes	0 Bytes		0.0%	11/13 (0)	Auto [Hi]	Wait
[eBook] P2P - Peer-to-Peer Computing.pdf	524 KB	60 KB	60 KB	4.0 kB/s	11.5%	7+2 (1)	Auto [Hi]	Dow
Infected_Mushroom_-_Cities_Of_The_Future-CDM-2004-MYCEL.(www.p...	37.11 MB	0 Bytes	0 Bytes		0.0%	13/83+16 (0)	Auto [Hi]	Wait
Convergence Of Peer-To-Peer And Grid Computing.pdf	62 KB	0 Bytes	0 Bytes		0.0%	0/5+1 (0)	Auto [Hi]	Wait
Hungry Minds - Peer-to-Peer Development and Production.pdf	2.40 MB	0 Bytes	0 Bytes		0.0%	0/5 (0)	Auto [Hi]	Wait
Peer to Peer- Harnessing the Power of Disruptive Technologies.pdf	2.12 MB	0 Bytes	0 Bytes		0.0%	0/8+4 (0)	Auto [Hi]	Wait
P2P Peer-To-Peer Architecture For Distributed Knowledge Management.ppt	93 KB	0 Bytes	0 Bytes		0.0%	0/5 (0)	Auto [Hi]	Wait
Infected Mushroom - Im The Supervisor-2004-Fwyh.rar	106.58 MB	0 Bytes	0 Bytes		0.0%	1/45+3 (0)	Auto [No]	Wait
Infected Mushroom (with Holymen) (goa - psy trance) - Desert Storm.mp3	7.26 MB	0 Bytes	0 Bytes		0.0%	12/51 (0)	Auto [No]	Wait
Infected Mushroom - Deeply Disturbed (2003).rar	42.75 MB	40 KB	40 KB	6.5 kB/s	0.1%	5/125+12 (1)	Auto [Lo]	Dow
Infected_Mushroom_-_Converting_Vegetarians_Retail-2CD-2003-IND.rar	209.45 MB	78 KB	66 KB	0.0 kB/s	0.0%	5/177+1 (1)	Auto [Lo]	Dow

Uploads (0)

Username	File	Speed	Transferred	Waited	Upload Time	Status
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Clients on queue: 0

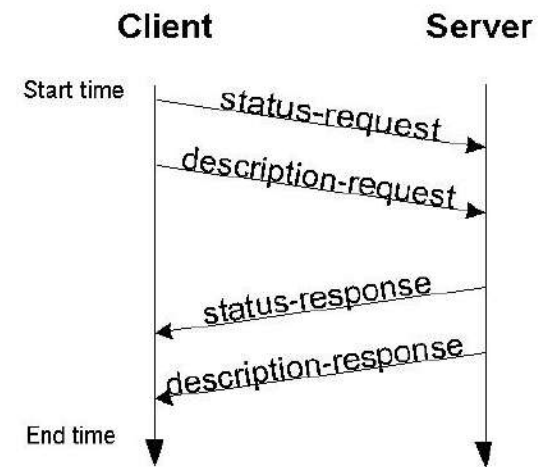
Connection established on:Razorback 2

Users: 492K(3.20M) | Files: 55.54M(354.62M) | Up: 0.0 | Down: 8.9

Razorback 2

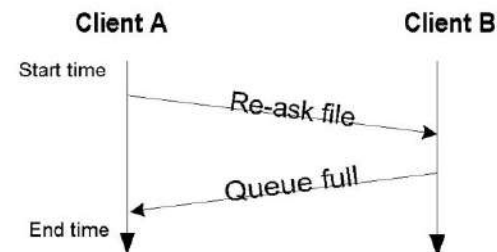
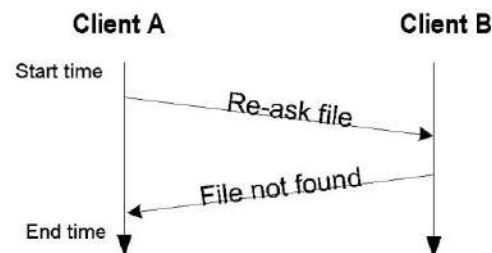
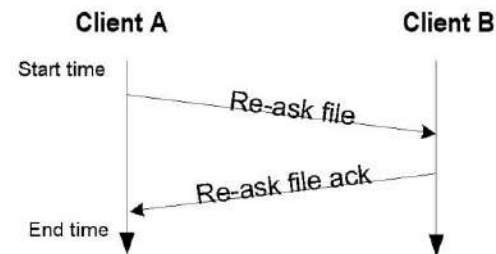
Client Server UDP communication

- UDP is used for:
 - periodic status checks
 - File search enhancement
 - Source search enhancement
- UDP is optional



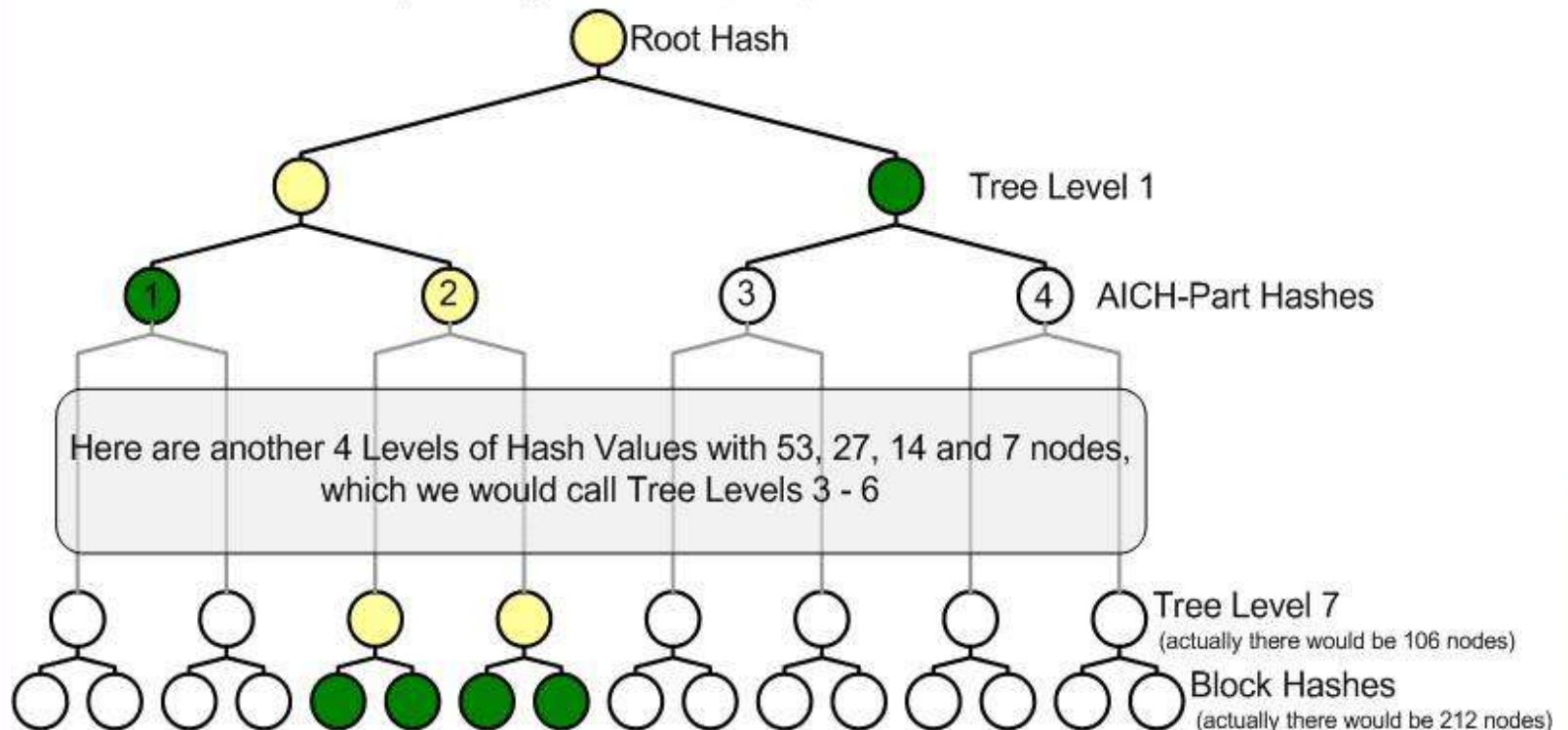
Client to Client UDP communication

- UDP is used for verifying A's position in B's upload queue



AICH

AICH Hash Tree for a 4 parts file
(Recovery Packet for part 2)



● Hashes inside the recovery packet: 53 Block Hashes + 2 Verifying Hashes

● Hashes which can be calculated from the green ones

emule-project.net

The image was taken from the eMule website

Free template from www.brainybetty.com